

RESEARCH METHODOLOGY & INTELLECTUAL PROPERTY RIGHTS

Course Code - AL58

Textbooks

1. C. R Kothari, Gourav Garg, Research Methodology – Methods and Techniques. New Age International Publishers.
2. Dr. B L Wadehra – Law relating to Intellectual property. Universal Law Publishing Co. Dipankar Deb, Rajeeb Dey, Valentina E. Balas
“Engineering Research Methodology”, ISSN 1868-4394 ISSN 1868-4408 (electronic), Intelligent Systems Reference Library, ISBN 978-981-13-2946-3 ISBN 978-981-13-2947-0 (eBook),
<https://doi.org/10.1007/978-981-13-2947-0>.

Reference Books

1. David V. Thiel “Research Methods for Engineers” Cambridge University Press, 978-1-107-03488-4

Course Assessment

- CIE- 30 Marks (Internal Exam 1 & 2)
- Research Publication or Indian Patent Publication – 10 Marks
- Numerical Test-10 Marks

Unit I

Introduction: Meaning of Research, Objectives of Research, Types of Research, Ethics in Research, Types of Research Misconduct.

(From C.R.Kothari Textbook)

Literature Review and Technical Reading: New and Existing Knowledge, Analysis and Synthesis of Prior Art, Bibliographic Databases, Conceptualizing Research, Critical and Creative Reading.

(From Dipankar Deb Rajeeb Dey Valentina E. Balas Textbook)

Citations: Functions and Attributes, Impact of Title and Keywords on Citations, Knowledge flow through Citations, Acknowledgments and Attributions.

(From Dipankar Deb Rajeeb Dey Valentina E. Balas Textbook)

Introduction to Research

Research Methodology

- Composed of two syllables “Re” and “Search”.
- “Re”=‘Again or over again or a new’ and
- “Search”=is the latter meaning ‘to examine closely and carefully’ or ‘to test and try’

Meaning of Research

- Search for **knowledge Scientific** and **systematic search** for information on a specific topic.
- An art of **scientific investigation** Systematized effort to gain new knowledge.
- A movement from the known to the unknown
It is actually a **discovery of a topic**



Research comprises

- **defining** and **redefining** problems,
- formulating hypothesis or suggested solutions;
- **collecting, organizing** and **evaluating data**;
- making deductions and **reaching conclusions**;
- at last carefully **testing the conclusions** to determine whether they fit the formulating hypothesis.

‘research’ refers

The systematic method consisting of enunciating the problem, formulating a hypothesis, collecting the facts or data, analyzing the facts and reaching certain conclusions either in the form of solutions(s) towards the concerned problem or in certain generalizations for some theoretical formulation.

OBJECTIVES OF RESEARCH

- **To gain familiarity with a phenomenon** or to achieve new insights into it.
- To portray accurately the **characteristics of a particular individual**, situation or a group.
- **To determine the frequency** with which something occurs or with which it is associated with something else.
- **To test a hypothesis** of a causal relationship between variables

MOTIVATION IN RESEARCH

- Desire to get a **research degree** along with its consequential benefits;
- Desire to **face the challenge** in solving the unsolved problems, i.e., concern over practical problems initiates research;
- Desire to get **intellectual joy** of doing some creative work;
- Desire to be of **service to society**;
- Desire to get **respectability**.

TYPES OF RESEARCH

Descriptive

- It includes surveys and fact-finding enquiries of different kinds.
- Researcher report what has happened or what is happening.
- Ex: Frequency of shopping, preference of people

Analytical

- Researcher has to use facts or information already available
- Analyze these to make a critical evaluation of the material.
- Ex: Explaining why and how US trade balance

TYPES OF RESEARCH

Applied

- Aims at finding a solution for an immediate problem facing a society, industrial organization.
- Ex: Development of medical treatments, Engineering innovations

Fundamental

- Concerned with generalizations and with the formulation of a theory.
- Ex: Study of factors impacting each stage of productive life cycle.

TYPES OF RESEARCH

Quantitative

- Its based on the measurement of quantity or amount.
- Applicable to fact/situation can be expressed in terms of quantity.
- Ex: amount of time a doctor takes to attend a patient when patient walks into the hospital

Qualitative

- Concerned with qualitative phenomenon i.e involving quality or kind.
- Ex: interested in investigating reasons for human behaviour.

TYPES OF RESEARCH

Conceptual

- Used philosophers & thinkers to develop new concepts.
- Researcher is conducted by observing & analyzing already present info on given topic
- Doesn't involve any practical experiment.
- Ex: Issac newton observation

Empirical

- Experimental type of research.
- Research based on observation, experiments & evidence.
- Ex: findings of DNA Testing

TYPES OF RESEARCH

Some Other Types of Research:

- **One-time research or longitudinal research.** In the former case the research is confined to a single time-period, whereas in the later case the research is carried on over several time-periods.
- **Field-setting research or laboratory research or simulation research,** depending upon the environment in which it is to be carried out.
- **Clinical or diagnostic research.** Such research follow case-study methods or in-depth approaches to reach the basic causal relations.
- **Conclusion-oriented and decision-oriented.** While doing conclusion-oriented research, a researcher is free to pick up a problem, redesign the enquiry as he proceeds and is prepared to conceptualize as he wishes.

Ethics in Research

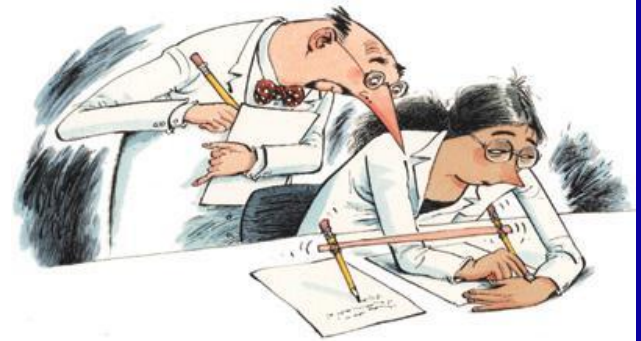
Provides guidelines for the responsible conduct of research.

Some ethical principles are

- Honesty
- Objectivity
- Integrity
- Carefulness
- Openness
- Respect for Intellectual Property
- Confidentiality

What are research misconducts?

Research Misconduct can be characterised as actions or questionable research practices that fall short of the standards of ethics, research and scholarship required to ensure that the integrity of research is upheld. It can cause harm to people and the environment, wastes resources, undermines the research record and damages the credibility of research.



TYPES OF RESEARCH Misconduct

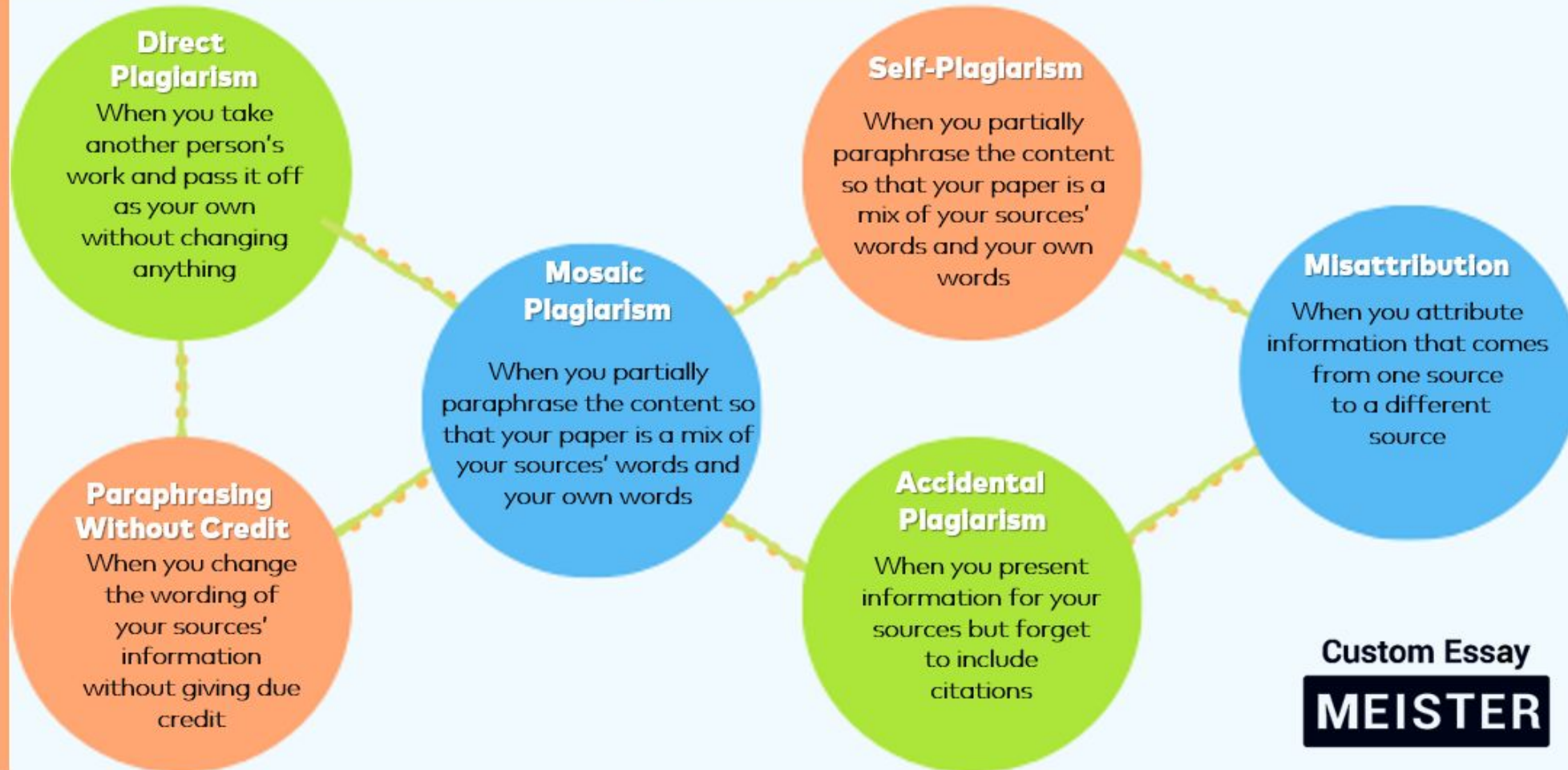
- **Fabrication** - making up data or results and recording or reporting them.
- **Falsification** - **manipulating research materials**, or changing or omitting data or results such that the research is not accurately represented in the research record. **Data errors** can also occur due to poor testing or inaccurate measurements.
- **Plagiarism** - appropriation of another person's ideas, processes or results without giving appropriate acknowledgment. Plagiarism is manifest when a person uses or reuses another work (including text, document, table, picture, diagram or content) as it appears in his work. Copying or reusing one's own published work is called self-plagiarism

How are supervisors, reviewers or editors alerted to plagiarism?

- (i) Original author comes to know and informs everyone concerned.
- (ii) Sometimes a reviewer finds out about it during the review process.
- (iii) readers who come across the article or book, while doing research.

Types of Plagiarism

Types of Plagiarism



Custom Essay
MEISTER

Literature Review and Technical Reading

Literature Review

The primary goal of a literature review is to:

Identify the **research problem**. This includes understanding the current state of knowledge on the topic, identifying gaps in knowledge, and determining the research questions that need to be answered.

Advocate a specific approach. This involves evaluating the different approaches that have been taken to study the problem and selecting the approach that is most likely to be successful.

Evaluate the choice of methods. This includes assessing the validity and reliability of the methods that have been used and determining whether they are appropriate for the research problem.

Demonstrate the need for new research. This involves showing that the existing research is not sufficient to answer the research questions and that new research is needed to make progress.

New and Existing Knowledge

- The **new knowledge** is often established by identifying existing problems in the field and demonstrating the gaps in the current understanding.
- **Existing knowledge** is essential for highlighting the existence of a problem and its importance. Researchers use the context, significance, originality, and tools from existing literature to build a case for their work..

- provides a strong foundation for advancing knowledge, identifying gaps, and suggesting new research directions.
A good literature survey is concept-focused rather than author-focused.

Components of a Comprehensive Literature Review:

- 1. Summarize existing knowledge** from the state of the art.
- 2. Detail key concepts**, factors, parameters & relationships.
- 3. Discuss complementary approaches** to the topic.
- 4. Highlight** inconsistencies, shortcomings, and contradictory **results**.
- 5. Justify the need** for further research in the field.

Steps to Conduct an Effective Literature Survey:

- 1. Identify major topics** or concepts relevant to the research subject.
- 2. Categorize relevant sources** (articles, patents, websites, data, etc.) under respective concepts

Analysis and Synthesis of Prior Art

The goal is to identify unsolved issues, flaws in existing models, and propose novel ideas.

A researcher should analyze the following steps:

- 1. Understanding the hypothesis:** Grasp the central research question or hypothesis of each article. This helps establish the context and purpose of the research

2. Understanding the models and the experimental conditions: Delve into the models and experimental setups used in the articles. This understanding helps in comparing and contrasting findings and methodologies

3. Making connections: Identify common themes, similarities, and differences across the articles. This step involves synthesizing the information to derive meaningful insights

4. Comparing and contrasting: Compare various pieces of information, methodologies, and results across different sources. This comparison can reveal trends, contradictions, or gaps in the existing knowledge.

5. Finding out the strong points and the loopholes: Evaluate the strengths and weaknesses of each article. Identify areas where the research is robust and where there might be limitations or areas that need further investigation.

Bibliographic Databases

- **“Bibliographic databases”** refer to **“abstracting and indexing services”** useful for collecting citation-related information and possibly abstracts of research articles from scholarly literature and making them available through search.
- A researcher should be able to quickly identify the databases that are of use in the idea or problem that one wishes to explore.
 - **Web of Science**
 - **Google and Google Scholar**

Web of Science

- It includes multiple databases, as well as specialized tools.
- It is a **good search tool** for scholarly materials requiring institutional license and **allows the researcher to search in a particular topic of interest** by selection in fields that are available in drop down menu such as title, topic, author, address, etc.
- The tool also allows sorting by number of citations (highest to lowest), publication date.

Google and Google Scholar

- (i) It's a **“black box”** of information. It searches everything on the Internet, with no quality control
- (ii) There are **limited search** functionality and refinement options.

limitations of Google Scholar :

1. Some of the results are not actually scholarly. **An article may look scholarly** at first glance, but is not a good source upon further inspection.

2. It is **not comprehensive**. Some publishers do not make their content available to Google Scholar.

3. There is limited search functionality and refinement options.

There are search operators that can be used to search relevant content:

(i) OR—Broadens search by capturing synonyms

Ex: Synchronous OR asynchronous will find results

(ii) Brackets()—Gather OR'd synonyms of a concept together, while combining them with another concept. **Ex:** RAM (synchronous OR asynchronous).

(iii) Quotation marks “ ”— search by finding words together as a phrase,

Ex: RAM (synchronous OR asynchronous) “Texas Instruments”.

(iv) Site—limits the search to results from a specific domain or website.

Ex: RAM (synchronous OR asynchronous) “Texas Instruments”
site: <http://ieeexplore.ieee.org>.

(v) Filetype—search with a specific file extension pdf’s, PPT’s, Excel sheets

Ex: RAM (synchronous OR asynchronous) “Texas Instruments”
site: <http://ieeexplore.ieee.org>, filetype: pdf.

Benefits of Bibliographic Databases

1. Abstracting and Indexing Services: Bibliographic databases include information such as citations, abstracts, authors, affiliations and keywords.

2. Access to Scholarly Literature: These databases facilitate access to a wide range of scholarly research articles, helping researchers stay informed about the latest developments in their field.

3. Search Capabilities: Researchers can perform advanced searches using keywords, authors, publication years, and other criteria to find relevant articles quickly.

4. Comprehensive Coverage: Bibliographic databases cover a vast array of disciplines and subjects, providing a comprehensive repository of research materials.

5. Quality Research: Simultaneous searches across multiple databases help researchers avoid overreliance on a single source and mitigate limitations associated with individual databases.

Conceptualizing Research

one needs to be continually reading the literature so as to bring together the three parts:

- (i) significant problem,** (the existing recorded literature and knowledge in the field.)
- (ii) the knowledge that will address it** (one is probably already an expert at the edge of knowledge else it is difficult to say with confidence)
- (iii) Possible way to make that new knowledge.**(requirement already asks us to think about the method and the tools)

Good research objective, conceptualizing the research that meets all requirements :

- one must already be aware of what is in the literature.
- one actually has a good research objective.
- one is probably already an expert at the edge of knowledge

Critical and Creative Reading

- Reading a research paper is a critical process.
- **Critical reading is relatively easy.**
 - It is relatively easier to critically read to **find the mistakes** than to read it so as to find the good ideas in the paper.
- In creative reading, the idea is to actively look for other **applications, interesting generalizations, or extended work** which the authors might have missed?
- Are there modifications that may throw up important practical challenges?



Citations

Citations: Functions and Attributes

- **Citations** (references) credit others for their work, while allowing the readers to trace the source publication if needed. the researcher may need to give credit to the creator of the original source.
- Citations help the readers to **verify the quality and importance of the new work** and **justification of the findings**.

- It is a way to tell readers that certain material in the researcher's present work has come from **another source** and credit has been given to the **original author**
- **A researcher needs to cite each source twice:**
 - In-text citation, in the text of the article exactly where the source is quoted or paraphrased,
 - a second time in the references, typically at the end of the chapter or a book or at the end of a research article.

- **LaTeX**, a document preparation system often used by engineering researchers to automatically format documents with standard formatting needs, is very effective to track and update citations.

Citations: Functions and Attributes

There are 3 main functions of citation:

1. **Verification function:** Authors have a scope for finding intentional or unintentional distortion of research or misleading statements in research. **Readers can verify the original source and determine if the cited claims are accurately represented in the current work.**

2. Acknowledgment function: Researchers primarily receive credit for their work through citations. Many reputed organizations and institutes provide research funding based on the reputations of the researchers. **Citations helps to build a researcher's reputation and provide valuable context for their work.**

3. Documentation function: Citations are also used to document scientific concepts and historical progress of any particular technology over the years.

Citations: Functions and Attributes

- There are certain cases when references do not fulfill the actual goal of citations and acknowledgments, and thus do not benefit the reader.
 - Spurious citations
 - Biased citations
 - Self-citations
 - Coercive citations

Spurious citations

- It occurs when authors include unnecessary or irrelevant citations. These citations waste readers' or reviewers' time and fail to enhance understanding of the paper.
- To maintain the credibility of the research and its publication, authors should avoid giving inappropriate credit.

Biased citations

- When authors cite friends or colleagues without genuine relevance, or intentionally omit significant works to avoid giving credit, especially if those works contradict their own.
- Such practices reflect citation bias, compromising the integrity of academic work.

Self citations

- The passage explains that self-citation is appropriate when it genuinely relates to the current research, as it can provide context within an ongoing project and help readers access related work.
- However, self-citation should be ethical and only include truly relevant prior works to avoid misleading the reader.

Coercive citations

- despite their limitations, impact factors are still widely used to measure research impact.
- However, this reliance creates incentives for editors to pressure authors into citing the journal to boost its impact factor, often implicitly suggesting that non-compliance could risk rejection. Such practices harm the journal's reputation.

Impact of Title and Keywords on Citations

- The **citation rate** of any research paper depends on various factors including significance and **availability of the journal, publication types, research area, and importance** of the published research work.
- Other factors like length of the **title**, type of the title, and **selected keywords also impact the citation count**

Title

- **Title** is the most important attribute of research paper.
- Title plays important role in **marketing** and makes research papers **traceable**.
- A good title is **informative**, represents a paper **effectively to readers**, and gains their attention.
- There are 3 different aspects which provide a particular behavior to the title:
 - **types of the title,** - **length of the title,**
 - **presence of specific markers**

Keywords

- **Keywords** represent essential information as well as main content of article, which are relevant to area of research.
- **Search engines, journal, digital libraries, and indexing services** use keywords for categorization of the research topic and to direct the work to the relevant audience.
- **Keywords** are important to ensure that readers are aware about research articles and their content.

Knowledge Flow Through Citation

- **Knowledge flows through verbal communications, books, documents, video, audio, and images, which plays a powerful role in research community**
- In engineering research, **knowledge flow** is primarily in the form of **books, thesis, articles, patents, and reports.**
- **Citing a source** is important for transmission of knowledge from previous work to an innovation.

- **Production of knowledge** can be related to the **citation network**.
- Knowledge flow happens **between co-authors during research collaboration**, among other researchers through their paper citation network, and also **between institutions, departments, research fields or topics**, and elements of research.

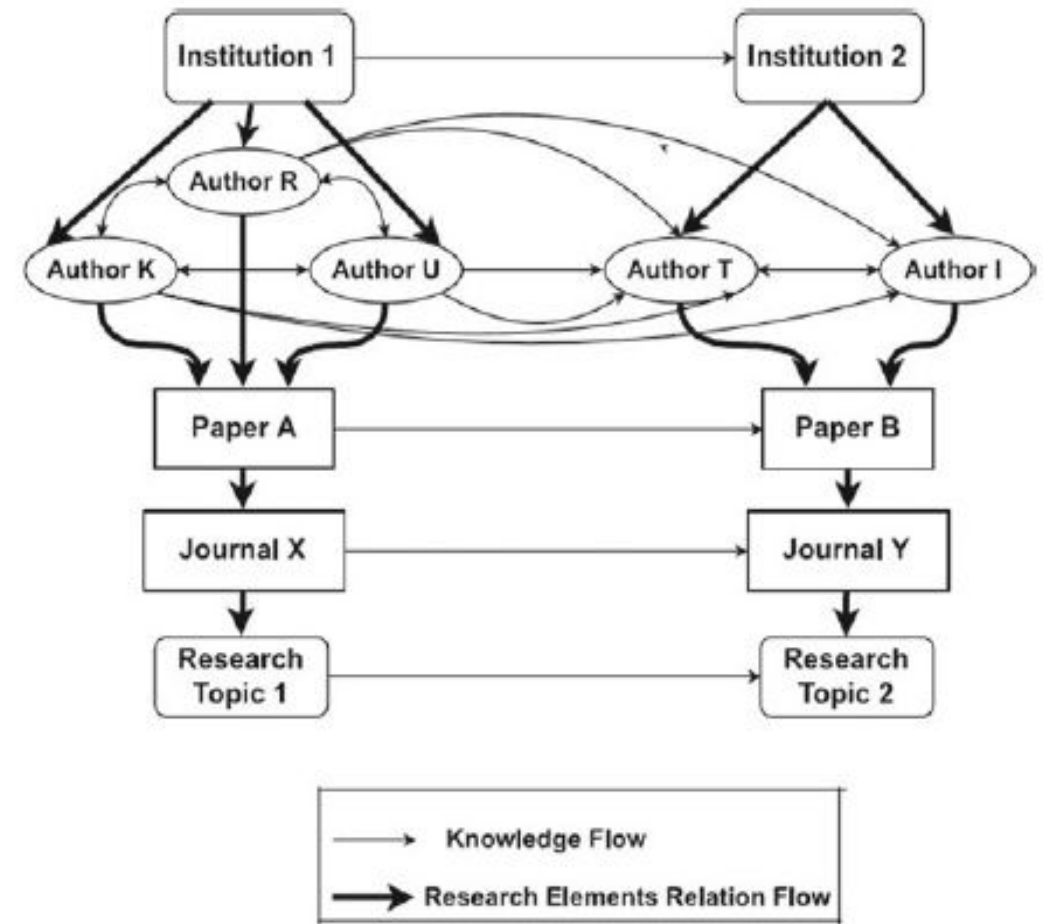


Fig : relationship between co-authorship and different types of citations

Citing Datasets

- **Data citations** should have provisions to give **credit and legal attribution** to all contributors, enable identification and access.
- **Researchers** should secure permission to use data and ensure dataset citations contain **sufficient information** for future retrieval, even if links become inactive.
- **Citations** should combine general and specific details to help readers reliably identify the dataset.

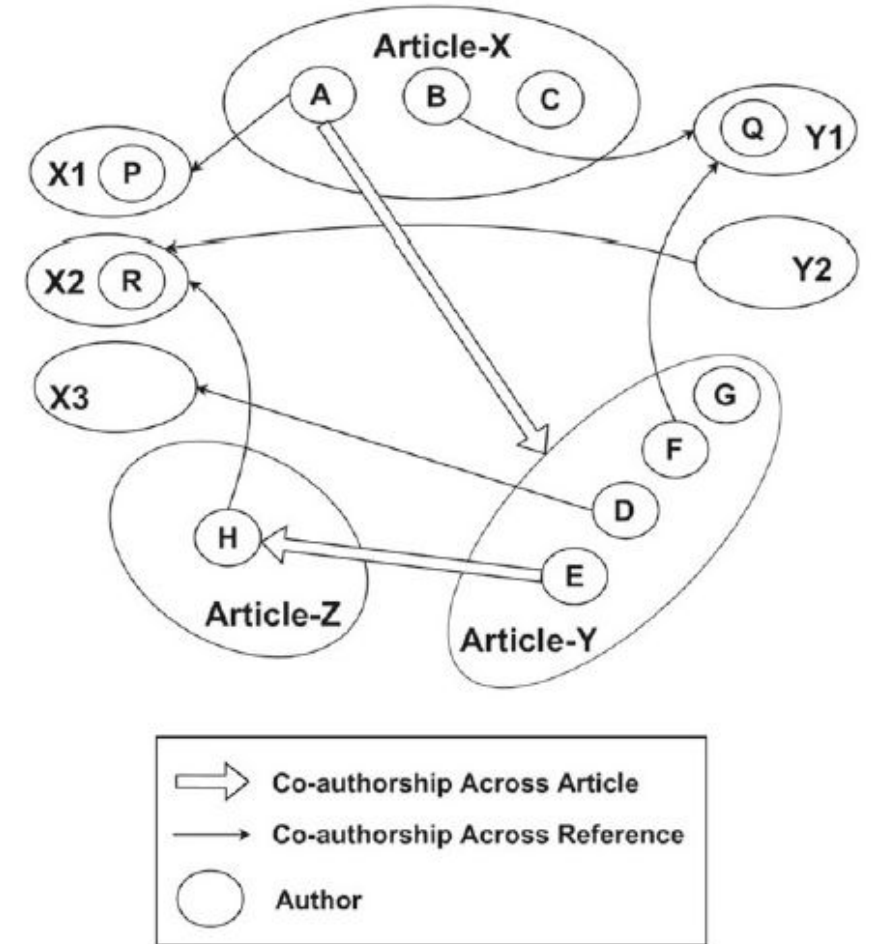


Fig: Co-authorship citation network

Styles for Citations

1. **ASCE style (American Society of Civil Engineers)**

- a) **Reference list:** this part is to be placed in the bibliography or references at the end of the report.
A template with example for the same is given below:

Template for books:

Author Surname, Author Initial. (Year Published). Title. Publisher, City, Pages Used.

Example:

Wearstler, K., and Bogart, J. (2004). Modern glamour. Regan Books, NY.

Template for websites:

Author Credentials / Company Name (Year Published). 'Title'. [http://Website URL](http://WebsiteURL) (Oct. 10, 2013).

Example:

Blade cleaning services (2015): <http://www.bladecleaning.com/problematica> (29 Oct, 2016).

Template for journal publications:

Author Surname, Author Initial. (Year Published). 'Title'. Publication Title, Volume number(Issue number), Pages Used.

Example:

Johnston, L. (2014). "How an Inconvenient Truth Expanded The Climate Change Dialogue abd Reignited An Ethical Purpose in The United States". 1-160.

b) In-text citation for journals or books: The following part is to be placed right after the reference to the source of the citation assignment:

Template

(Author Surname/Website URL Year Published)

Examples:

- i. Citation is a very important part of technical writing. (Deb 2016)
- ii. Engineers create devices to monitor mountains so that nearby inhabitants can be warned of impending eruptions. (Teachengineering.org 2014)

2. IEEE style (Institute of Electrical and Electronics Engineers) IEEE style is standard for all IEEE journals and magazines, and is frequently used for papers and articles in the fields of electrical engineering and computer science.

Chapter in an edited book

[1] A. Rezi and M. Allam, "Techniques in array processing by means of transformations," in Control and Dynamic Systems, Vol. 69, Multidimensional Systems, C. T. Leondes, Ed. San Diego: Academic Press, 1995, pp. 133–180.

3. ASME style (The Association of Mechanical Engineers)

4. MLA (Modern Language Association)

- Used mostly in **arts, humanities, literature**. It Focuses on **author names + title + publisher + year**.
- **Ex:**

MLA Natikar, Kavya, and R. B. Dayananda. "Addressing Crop Damage from Animals with Faster R-CNN and YOLO Models." *International Conference on IoT Based Control Networks and Intelligent Systems*. Singapore: Springer Nature Singapore, 2023.

5. APA (American Psychological Association)

- Common in **social sciences, psychology, education**.
It Focuses on **author initials + year in brackets + month**
- **Ex:**

APA Natikar, K., & Dayananda, R. B. (2023, June). Addressing Crop Damage from Animals with Faster R-CNN and YOLO Models. In *International Conference on IoT Based Control Networks and Intelligent Systems* (pp. 651-666). Singapore: Springer Nature Singapore.

6. Chicago

- Used in **history, business, publishing.**
- Gives **full details, similar to MLA, but includes page numbers clearly.**
- **Ex:**

Chicago Natikar, Kavya, and R. B. Dayananda. "Addressing Crop Damage from Animals with Faster R-CNN and YOLO Models." In *International Conference on IoT Based Control Networks and Intelligent Systems*, pp. 651-666. Singapore: Springer Nature Singapore, 2023.

7. Harvard

- Very common in **universities, research projects.**
- Looks like APA but **simpler** – author, year, title, pages.
- **Ex:**

Harvard Natikar, K. and Dayananda, R.B., 2023, June. Addressing Crop Damage from Animals with Faster R-CNN and YOLO Models. In *International Conference on IoT Based Control Networks and Intelligent Systems* (pp. 651-666). Singapore: Springer Nature Singapore.

8. Vancouver

- Mostly used in **medical and scientific journals**.
- Very short, **no quotation marks, initials after surname, date at the end.**
- **Ex:**

Vancouver Natikar K, Dayananda RB. Addressing Crop Damage from Animals with Faster R-CNN and YOLO Models. In International Conference on IoT Based Control Networks and Intelligent Systems 2023 Jun 21 (pp. 651-666). Singapore: Springer Nature Singapore.

Style	Used in	Main Features
MLA (Modern Language Association)	Humanities, literature	Author full name, title in quotes, publisher & year
APA (American Psychological Association)	Social sciences, education, psychology	Author initials, year in brackets, clear date, page numbers
Chicago	History, business, publishing	Similar to MLA but emphasizes publication year & pages
Harvard	Widely used in universities & research	Author–year style, simpler than APA
Vancouver	Medical & scientific journals	Surname + initials, minimal punctuation, date at end

Acknowledgments and Attributions

- **Acknowledgment section** is a place to provide a brief **appreciation of the contribution** of someone or an organization or funding body to the present work.
- If no particular guideline is available for the intended publication, then **it can be introduced at the end of the text** or as a footnote.

- **Acknowledgment** is a common practice to **recognize persons** or agencies for being responsible in some form or other for completion of a publishable research outcome.
- Acknowledgment displays a **relationship among people, agencies, institutions, and research.**

What Should Be Acknowledged?

- Every author should know that what **should/should not be acknowledged**.
- Author should acknowledge **quotation, ideas, facts, paraphrasing, funding organization or support, laboratory, and computer work**.

Acknowledgments:

This research work was funded in part by the Extra Mural Research Funding 2014–17 (Individual Centric) of the Department of Science and Technology (DST), Govt. of India.

Acknowledgments in Books/Dissertations

- A page of acknowledgments is usually included at the **beginning of a thesis**/dissertation immediately following the table of contents.

Sample Acknowledgement in Thesis:

I wish to express my sincere appreciation to my supervisor Prof. Gang Tao for the useful comments, remarks and encouragement throughout this thesis work. Furthermore, I wish to express my thanks to Prof. Jacob Hammer for introducing me to the topic and for the support along the way. Also, I like to thank my peers in the Adaptive Control Lab such as Yu Liu and Shan-shan Li, who have shared their precious time during many lively technical discussions. I would like to thank my family members who have supported me throughout this journey in many different ways.

Dedication or Acknowledgments?

- **Dedication** is almost never used in a journal paper, an article in a conference proceedings, or a patent, and it is used in larger documents like **books, thesis, or dissertations**.
- While **acknowledgments** are reserved for those who helped out with the book in some way or another (editing, moral support, etc.), a **dedication** is to whomever the author would like it to be dedicated to, whether it is the author's parents, Spouse, Children, the best friend, the pet dog, or Almighty God.

"Research is creating new knowledge."

